Lecture Notes on Convex Optimization and Optimal Control

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1 Introduction

Optimization and optimal control study decision-making problems in which one seeks to choose variables, functions, or control inputs so as to optimize a specified performance criterion subject to constraints. These problems arise throughout engineering, economics, operations research, and the natural sciences, where limited resources and competing objectives must be balanced in a principled manner.

At an abstract level, an optimization problem can be formulated as

$$\min_{x \in C} f(x), \tag{1.1}$$

where $f : C \to \mathbb{R}$ is the *objective function* and C is the *feasible (or constraint) set*. The objective function quantifies performance, cost, or utility, while the constraint set encodes admissibility requirements on the decision variable x. This formulation implicitly assumes that objective values can be compared via a total order, here induced by the usual ordering on $\mathbb{R}$.

The use of the notation min further assumes that an optimal solution exists and that the minimum is attained; when this is not guaranteed, the problem is more precisely described in terms of an infimum.

An element $x^\star \in C$ is called an *optimal solution* (or *minimizer*) if

$$f(x^\star) \leq f(x), \qquad \forall x \in C. \tag{1.2}$$

That is, an optimal solution achieves the smallest objective value among all feasible decisions.

The set of all optimal solutions is denoted by

for whichever $y \in C$ $f(y) \approx \infty$

set →

$$\mathcal{X}^\star := \{x \in C \mid f(x) = \inf_{y \in C} f(y)\}, \tag{1.3}$$

and may be empty, contain a single element, or consist of multiple solutions, depending on the structure of f and C.

The *optimal value* of the problem is defined as

$$f^\star := \inf_{x \in C} f(x), \tag{1.4}$$

which represents the best achievable performance according to the objective function. When there exists $x^\star \in C$ such that $f(x^\star) = f^\star$, the infimum is said to be *attained*, and $x^\star$ is an optimal solution.

All definitions extend analogously to maximization problems of the form

$$\max_{x \in C} f(x),$$

by replacing inf with sup and reversing the inequality in the definition of optimality.

These basic concepts provide the foundational language for optimization and optimal control. Subsequent sections study the conditions under which optimal solutions exist, whether they are unique, how they can be characterized, and how they can be computed or approximated in practice.

1.1 Finite- and Infinite-Dimensional Formulations

In finite-dimensional optimization, the decision variable x belongs to $\mathbb{R}^n$, and the objective function is a mapping

$$f : \mathbb{R}^n \to \mathbb{R}, \tag{1.5}$$

with $C \subseteq \mathbb{R}^n$. Under suitable assumptions on f and C, one may characterize optimal solutions using tools such as first- and second-order optimality conditions, duality theory, and iterative algorithms.

In many applications, however, the decision variable is not a finite-dimensional vector but a function. In such cases, the objective is a *functional*

$$f : V \to \mathbb{R}, \tag{1.6}$$

defined on an infinite-dimensional vector space V. A common example is

$$V = C^k([a,b], \mathbb{R}^n), \qquad k \geq 0, \tag{1.7}$$

the space of k-times continuously differentiable functions from the interval $[a,b]$ into $\mathbb{R}^n$.

In this infinite-dimensional setting, optimization problems are studied within the calculus of variations and optimal control theory. The decision variable represents a trajectory or control signal evolving over time, and optimality conditions are expressed in terms of differential equations and variational arguments rather than finite-dimensional gradients.

The feasible set C may impose a variety of constraints, including pointwise bounds, integral constraints, boundary conditions, or differential equations that govern admissible trajectories. In optimal control problems, these differential constraints typically represent the system dynamics.

A basic example of a functional optimization problem is

$$f(x) = \int_a^b g(x(t), t), dt, \tag{1.8}$$

where $g : \mathbb{R}^n \times [a,b] \to \mathbb{R}$ is a given integrand representing instantaneous cost or utility. The decision variable is a function $x(\cdot)$ defined on the interval $[a,b]$, and the objective evaluates the aggregate performance accumulated over time.

The admissible functions $x(\cdot)$ are typically restricted to a suitable function space, such as $C([a,b], \mathbb{R}^n)$ or $C^1([a,b], \mathbb{R}^n)$, and may be required to satisfy pointwise constraints of the form

$$|x(t)| \leq d, \qquad \forall t \in [a,b], \tag{1.9}$$

as well as boundary conditions such as $x(a) = x_a$ and $x(b) = x_b$. Problems of this type arise naturally in the calculus of variations and provide a direct generalization of finite-dimensional constrained optimization to function spaces.

Another class of problems arises when the admissible functions $x(\cdot)$ are required to satisfy differential constraints of the form

$$\dot{x}(t) = f(x(t), t), \qquad x(a) = x_a, \tag{1.10}$$

where f specifies the evolution of the state over time. In this case, the optimization problem

becomes

$$\min_{x(\cdot)} \int_a^b g(x(t), t), dt \quad \text{subject to} \quad \dot{x}(t) = f(x(t), t). \tag{1.11}$$

The differential equation restricts the feasible set to trajectories that are dynamically consistent with the prescribed evolution law.

Such problems form a bridge between the calculus of variations and optimal control theory. When the dynamics depend on an additional control input $u(\cdot)$, the problem naturally extends to a full optimal control formulation involving both state and control variables, with optimality characterized by conditions such as the Euler–Lagrange equations or Pontryagin's Maximum Principle.

1.2 Systems and Optimization

Optimization and optimal control are fundamentally *systems methods*. In this context, a *system* is an abstraction that characterizes how inputs are mapped to outputs, possibly through internal states, constraints, and interconnections. The defining feature of the systems viewpoint is the focus on *input–output behavior*: decisions are evaluated according to how they shape observable outcomes and aggregate performance, rather than by a complete description of internal mechanisms.

This systems viewpoint contrasts with a *non-system* or *mechanistic* approach, which is characteristic of many areas of physics and biology. In those disciplines, the primary objective is often to *understand what is inside the box*: to uncover the underlying structure, laws, or mechanisms that govern observed phenomena. Such approaches emphasize explanatory fidelity, causal decomposition, and internal consistency, even when the resulting models are too complex to support direct decision-making or control.

By contrast, systems methods deliberately abstract away internal detail in favor of externally relevant behavior. From a systems perspective, the internal structure of the box matters only insofar as it constrains or explains the input–output relationships of interest. This abstraction allows one to reason about performance, robustness, and trade-offs without requiring a complete or microscopically accurate model of the underlying processes.

Within this framework, optimization is not an end in itself, but a *tool for system analysis and design*. It enables the systematic study of which input–output behaviors are achievable under given constraints, the quantification of performance limits, and the synthesis of decision rules that shape system behavior toward desired objectives. Optimal control extends this framework

by embedding optimization within feedback architectures, allowing inputs to be adapted based on observed outputs and evolving system behavior.

Accordingly, optimization and optimal control problems are typically addressed through four closely related stages for a system: *modeling*, *analysis*, *design*, and *implementation*

1.2.1 System Modeling

The modeling stage concerns the abstraction of a real-world system into a mathematical formulation suitable for optimization. This abstraction requires identifying the essential components of the system—such as states, inputs, outputs, and disturbances—while deliberately omitting secondary details that are negligible, unobservable, or analytically intractable. The objective is not to replicate reality in full detail, but to capture the causal and structural relationships that govern system behavior and are most relevant to the control objective.

In optimization and optimal control, modeling typically involves three core elements. First, one specifies the *decision variables*, representing the degrees of freedom available to the decision-maker. These may be static variables, time-varying trajectories, or control inputs acting on a dynamical system. Second, one defines a *performance criterion* that quantifies desired system behavior, such as cost, energy expenditure, risk, or utility, often accumulated over time. Third, one introduces *constraints* that encode physical laws, resource limitations, safety requirements, or informational restrictions.

A central modeling consideration is the choice of an appropriate level of fidelity. High-fidelity system models may incorporate nonlinear dynamics, uncertainty, and detailed interactions, but can be analytically intractable or computationally prohibitive. Low-fidelity models may sacrifice accuracy in favor of analytical clarity and insight. Effective systems modeling strikes a balance between these extremes, preserving qualitative structure while enabling rigorous analysis and tractable solution methods.

1.2.2 System Analysis

Once a system model has been specified, optimization provides a framework for *system analysis*: the study of fundamental performance limits and structural properties implied by the model. From a systems viewpoint, analysis seeks to determine what input–output behaviors are achievable under the given constraints, how performance depends on system parameters, and what trade-offs are inherent in the system architecture. Importantly, system analysis focuses on properties that are intrinsic to the model and therefore independent of any specific

implementation or algorithm.

At a mathematical level, system analysis gives rise to a set of foundational questions that underlie any optimization problem.

Existence. The existence question concerns whether the optimization problem admits at least one optimal solution. In finite-dimensional settings, existence is often established using compactness and continuity arguments, such as the Weierstrass extreme value theorem. In infinite-dimensional or optimal control problems, existence typically requires additional structure, such as coercivity, convexity, or weak compactness of the feasible set. Establishing existence is essential, as it ensures that the modeled system admits a well-defined notion of optimal behavior.

Uniqueness. When an optimal solution exists, one may ask whether it is unique. Uniqueness is typically guaranteed under strict convexity of the objective function and convexity of the feasible set. In optimal control problems, uniqueness may also depend on the system dynamics and terminal conditions. Non-uniqueness is common in practice and may reflect symmetries in the system, redundancy in the control inputs, or indifference among multiple optimal input–output behaviors.

Characterization. Characterization refers to identifying conditions that optimal solutions must satisfy. In finite-dimensional optimization, these conditions include first-order stationarity, second-order optimality conditions, and complementary slackness relations. In the calculus of variations and optimal control, characterization takes the form of differential equations and variational inequalities, such as the Euler–Lagrange equations, Pontryagin's Maximum Principle, and Hamilton–Jacobi–Bellman equations. These characterizations provide structural insight into optimal system behavior and form the basis for both theoretical understanding and algorithmic design.

Computation. The computation question concerns how optimal solutions can be obtained algorithmically once existence and characterization have been established. Only a limited class of optimization and optimal control problems admit closed-form solutions, such as linear–quadratic regulators or unconstrained convex programs. In general, optimal solutions must be computed numerically.

Classical computational approaches exploit problem structure and include gradient-based and Newton-type methods, interior-point and active-set methods for constrained optimiza-

tion, dynamic programming for sequential decision problems, and iterative schemes based on variational principles or fixed-point equations. In optimal control, computation often involves solving boundary-value problems, Riccati equations, or Hamilton–Jacobi–Bellman equations, either exactly or approximately.

In contemporary settings, computation increasingly incorporates *learning* and *data-driven methods*. When system models are incomplete, unknown, or too complex for direct analysis, data may be used to approximate value functions, policies, gradients, or system responses. Learning-based methods, including reinforcement learning, approximate dynamic programming, and policy optimization, can be viewed as computational tools for solving or approximating optimization and optimal control problems using observed data rather than explicit models.

The feasibility of computation therefore depends not only on problem dimension and structure, but also on data availability, sample efficiency, and the interaction between learning and optimization. Issues such as convergence, generalization, robustness, and computational complexity play a central role, particularly in large-scale, uncertain, or real-time environments. As a result, the computational question is no longer merely whether an optimal solution can be computed in principle, but whether it can be computed reliably, efficiently, and adaptively given limited information and data.

1.3 System Design and Implementation

System design and implementation translate the results of analysis and computation into concrete decision mechanisms that operate within a real system. While analysis characterizes what is achievable in principle and computation addresses how optimal solutions can be obtained algorithmically, design and implementation concern how these solutions are realized as operational system components.

System Design. From a systems perspective, design is the process of synthesizing *decision rules* that shape system behavior in accordance with specified performance objectives. In optimization and optimal control, this typically takes the form of policies, controllers, or decision mappings that assign inputs based on available information, such as system states, outputs, or measurements.

Design is inherently constrained by the system abstraction. Information availability, actuation limits, communication constraints, and uncertainty all restrict the class of admissible deci-

sion rules. As a result, system design often involves structural choices, such as selecting feedback versus open-loop architectures, centralized versus distributed control, or model-based versus data-driven decision mechanisms. These choices determine how optimization results are embedded into the system and how decisions interact with the system's input–output behavior.

Importantly, design does not necessarily require exact optimality. In many applications, the objective is to achieve satisfactory or robust performance under uncertainty rather than strict optimality with respect to an idealized model. Approximate, hierarchical, or receding-horizon designs are therefore common and reflect practical trade-offs between performance, complexity, and robustness.

Implementation. Implementation concerns the realization of designed decision rules in real systems. This stage accounts for non-idealities that are not captured fully in the abstract model, including finite computation time, numerical precision, delays, noise, incomplete information, and model mismatch. From a systems viewpoint, implementation is not a purely technical afterthought, but a critical component that influences achievable input–output behavior.

In optimal control and learning-based systems, implementation may involve real-time computation, online adaptation, or continual data acquisition. Constraints on memory, computation, and communication can significantly affect how optimization-based decisions are executed. Consequently, implementation considerations often feed back into modeling and design, motivating simplified models, reduced-order representations, or approximate solution methods.

Design–Implementation Interaction. System design and implementation are rarely sequential and are best viewed as part of an iterative loop. Limitations encountered during implementation may necessitate revisions to the system model, performance criteria, or control architecture. Conversely, advances in computation, sensing, and learning can enable more sophisticated designs and richer system abstractions.

1.4 Descriptive and Prescriptive Roles of Optimization.

A useful distinction in systems and decision sciences is between *descriptive* and *prescriptive* approaches. Descriptive methods aim to model, explain, or predict how a system behaves, whereas prescriptive methods aim to determine how a system *should* behave in order to achieve

specified objectives. Descriptive approaches are common in disciplines such as physics, biology, and statistics, where the primary goal is to understand the input–output behavior of a system as it is observed in nature. In this setting, models are evaluated based on their explanatory power, predictive accuracy, or consistency with empirical data. Learning and data-driven techniques are often used in a descriptive role to identify system dynamics, estimate parameters, or forecast future outputs given observed inputs.

While optimization and optimal control are often associated with *prescriptive* decision-making, optimization also plays an important *descriptive* role in the modeling and understanding of systems. In a descriptive role, optimization is used to explain or predict observed system behavior by positing that the system evolves so as to optimize an implicit objective. Classical examples include physical systems characterized by energy minimization principles, biological systems modeled through fitness maximization, and economic systems described by utility optimization. In these settings, the optimization problem is not a control prescription imposed by an external decision-maker, but a modeling hypothesis that captures the governing input–output behavior of the system. When valid, such models provide explanatory insight and predictive power.

In its prescriptive role, optimization is used to determine how inputs or decisions *should* be chosen to achieve desired outcomes under specified constraints. Here, the objective function reflects design goals rather than emergent system tendencies, and optimization serves as a normative tool for synthesizing decision rules or control policies.

From a systems perspective, the distinction lies not in the mathematical form of the optimization problem, but in its interpretation. In descriptive optimization, the objective function encodes hypothesized system principles that explain observed behavior. In prescriptive optimization, the objective function encodes desired performance criteria that guide intervention and design. Modern learning-based methods often blur this boundary: data may be used to infer implicit objectives or value functions descriptively, and those same constructs may then be used prescriptively to shape future system behavior.

2 Examples

2.1 Resource Allocation

A canonical class of optimization problems arises in *resource allocation*, where a limited resource must be distributed among multiple agents or activities. Such problems provide a simple yet

powerful illustration of how objective functions, feasible sets, and ordering assumptions enter an optimization model.

Problem formulation. Consider N agents sharing a divisible resource of total capacity $C > 0$. Agent i receives an allocation $x_i \geq 0$ and derives utility $f_i(x_i)$ from that allocation. A central planner seeks to allocate the resource so as to maximize total utility:

$$\max_{x_1,\ldots,x_N} \quad \sum_{i=1}^{N} f_i(x_i) \text{ s.t.} \qquad \sum_{i=1}^{N} x_i \leq C,\, x_i \geq 0, \quad i = 1, \ldots, N. \tag{2.1}$$

Decision variables and feasible set. The decision variable is the allocation vector

$$x = (x_1, \ldots, x_N) \in \mathbb{R}^N.$$

The constraints define the feasible set

$$\mathcal{C} = \left\{ x \in \mathbb{R}^N \;\middle|\; x_i \geq 0 \text{ for all } i = 1, \ldots, N,\ \sum_{i=1}^{N} x_i \leq C \right\},$$

which captures both the nonnegativity requirements and the scarcity of the shared resource. In the abstract optimization problem

$$\min_{x \in C} f(x),$$

this set $\mathcal{C}$ plays the role of the constraint set C.

Objective function and ordering. The objective function is given by $f(x) := \sum_{i=1}^{N} f_i(x_i)$, which aggregates individual utilities into a scalar performance measure. Because $f : \mathcal{C} \to \mathbb{R}$ is real-valued, the usual total order on $\mathbb{R}$ induces a total preorder on feasible allocations. In particular, for any two feasible allocations $x, y \in \mathcal{C}$, either $f(x) \leq f(y)$ or $f(y) \leq f(x)$ holds. needs clarification This comparability allows feasible allocations to be meaningfully ranked and enables the formulation of a well-defined optimization problem.

The abstract utility function f_i admits concrete interpretations depending on the application domain. In each case, f_i maps an individual resource allocation x_i to a scalar measure of performance or benefit, thereby allowing heterogeneous preferences to be aggregated into a single system-level objective.

Congestion control . Here x_i denotes the bandwidth allocated to user i in a communication network. A widely used choice is

$$f_i(x_i) = \log(x_i + 1),$$

which reflects diminishing marginal utility of bandwidth and leads to *proportional fairness*. Under this objective, any feasible reallocation that increases one user's bandwidth by a given percentage must decrease the aggregate bandwidth of others by at least the same percentage. This choice of f_i yields tractable convex optimization problems and aligns well with observed behavior in distributed congestion-control protocols.

Portfolio optimization . In financial settings, x_i represents the capital invested in asset i. A common quadratic utility model is

$$f_i(x_i) = \mu_i x_i - \tfrac{1}{2}\sigma_i^2 x_i^2,$$

where μ_i is the expected return and σ_i^2 quantifies risk or variance. The linear term rewards expected gain, while the quadratic penalty captures risk aversion. Summing such utilities across assets leads to classical mean–variance portfolio optimization, where the objective encodes a trade-off between return and risk in a single scalar criterion.

Urban planning. In urban planning problems, the allocation variable x_i typically represents a limited resource such as budget, land area, housing units, or infrastructure capacity assigned to project or district i. The function $f_i(x_i)$ models the *social benefit* or *economic value* generated by that allocation. The aggregate objective $f(x) = \sum_{i=1}^{N} f_i(x_i)$ represents total social welfare, economic output, or quality-of-life metrics across the city or region.

Scheduling. In scheduling problems, the allocation variable x_i typically represents processing time, machine capacity, or computational resources assigned to task i. The function $f_i(x_i)$ measures the *value* of allocating that amount of time or capacity to the task.

Typical formulations include:

- *Completion-value models*, such as

$$f_i(x_i) = v_i \min\{x_i, d_i\},$$

where d_i denotes the processing time required for completion and v_i is the priority or reward associated with task i.

- *Delay-penalty models*, for example

$$f_i(x_i) = -c_i \max\{d_i - x_i, 0\},$$

which penalize insufficient allocation through tardiness or deadline violations.

- *Concave utility models*, such as

$$f_i(x_i) = w_i \log(1 + x_i),$$

capturing diminishing returns from allocating processing time beyond a core requirement.

In this setting, maximizing

$$\sum_{i=1}^{N} f_i(x_i)$$

corresponds to prioritizing tasks, balancing throughput and fairness, or minimizing aggregate delay costs.

Although the interpretations differ across domains, the mathematical role of f_i is the same: it maps an individual allocation to a scalar measure of value. By aggregating these functions, the planner induces a total order over feasible allocations, enabling comparison, ranking, and optimization within a common framework.

From a *prescriptive* perspective, problem (2.1) specifies how a planner *should* allocate resources to achieve efficiency, fairness, or other design objectives. From a *descriptive* perspective, similar optimization problems are used to explain observed allocation patterns by hypothesizing that systems evolve toward configurations that maximize aggregate utility.

This example illustrates how concrete modeling choices give operational meaning to the abstract components of optimization: the feasible set $\mathcal{C}$ encodes system constraints, the objective function f induces an ordering over allocations, and optimal solutions represent preferred system-level outcomes.

2.2 Shortest-Time (Brachistochrone-Type) Problem

A classical example illustrating optimization over *functions* rather than finite-dimensional vectors is the *brachistochrone problem*, which originates in the calculus of variations and predates modern optimal control.

Physical setting. Consider a particle sliding without friction under uniform gravity from a starting point a to an endpoint b in a vertical plane. The particle is constrained to move along a curve $y = y(x)$ connecting these two points, where the vertical coordinate y is measured downward from the initial height so that $y(x) > 0$ along the trajectory. Among all admissible curves satisfying the boundary conditions

$$y(a) = y_a, \qquad y(b) = y_b,$$

the objective is to determine the curve along which the particle reaches b in the shortest possible time.

Modeling and objective functional. Under conservation of energy, the speed of the particle at height y is proportional to $\sqrt{y}$. The infinitesimal time required to traverse an arc-length element ds is therefore proportional to

$$\frac{ds}{\sqrt{y}}.$$

Since the arc length satisfies

$$ds = \sqrt{1 + \big(y'(x)\big)^2}\, dx,$$

the total travel time along a curve $y(\cdot)$ can be written as the functional

$$J[y] = \int_a^b \frac{\sqrt{1 + \big(y'(x)\big)^2}}{\sqrt{y(x)}}\, dx = \int_a^b g\big(y(x), y'(x)\big)\, dx, \tag{2.2}$$

where

$$g(y, y') := \frac{\sqrt{1 + (y')^2}}{\sqrt{y}}.$$

Here, the *decision variable* is the function $y(\cdot)$ itself, the *feasible set* consists of sufficiently smooth curves satisfying the boundary conditions and the positivity constraint $y(x) > 0$, and the objective is to minimize the scalar functional $J[y]$. The integral induces a total order on admissible trajectories via their travel times.

Analysis via Euler–Lagrange equations. The problem (2.2) is an infinite-dimensional optimization problem. Necessary conditions for optimality are obtained by applying the Euler–Lagrange equation

$$\frac{d}{dx}\left(\frac{\partial g}{\partial y'}\right) - \frac{\partial g}{\partial y} = 0.$$

Exploiting the fact that g does not depend explicitly on x, one may further derive a first integral of motion, leading to a closed-form characterization of the optimal curve.

Solving the resulting equations shows that the minimizing trajectory is a *cycloid*, that is, the curve traced by a point on the rim of a rolling circle. Remarkably, this curve yields the shortest descent time regardless of the relative horizontal and vertical separation between the endpoints, provided the particle moves under gravity alone.

The brachistochrone problem illustrates several ideas that are central to optimal control and systems theory. First, the optimization is performed over entire trajectories rather than static decision variables, highlighting the infinite-dimensional nature of many control problems. Second, the performance criterion is expressed as an objective functional defined by an integral over space or time, a structure that recurs throughout the calculus of variations and optimal control. Third, the form of the cost functional is dictated by physical modeling principles, in this case, conservation of energy, showing how system dynamics shape optimization objectives. Finally, the problem demonstrates how simple underlying principles can give rise to highly nontrivial optimal solutions, such as the cycloid.

Historically, this problem marks one of the earliest instances where system behavior is explained and predicted by an underlying optimization principle. Conceptually, it provides a bridge between static optimization, the calculus of variations, and modern optimal control, where similar integral cost functionals arise naturally in dynamical systems and trajectory planning.

2.3 Epidemic Control Model

A canonical example of optimal control in dynamical systems arises in *epidemic control*, where intervention policies are designed to mitigate the spread of an infectious disease while accounting for limited resources and societal costs.

System dynamics. Consider a controlled susceptible–infected–removed (SIR) model given by

$$\dot{S}(t) = -\beta S(t)I(t), \tag{2.3}$$

$$\dot{I}(t) = \beta S(t)I(t) - u(t), \tag{2.4}$$

$$\dot{R}(t) = u(t), \tag{2.5}$$

where $S(t)$, $I(t)$, and $R(t)$ denote the susceptible, infected, and removed (population recovered, isolated, or vaccinated) compartments at time t, respectively. The parameter $\beta > 0$ represents the transmission rate of the disease. The control input $u(t)$ models an intervention such as treatment, isolation, or vaccination that transfers individuals from the infected to the removed compartment.

Decision variables and feasible controls. The decision variable in this problem is the control trajectory

$$u(\cdot) : [t_0, t_1] \to \mathbb{R}_{\geq 0},$$

which specifies the intervention intensity over time. Admissible controls are typically required to satisfy pointwise constraints

$$u(t) \in \mathcal{U} := [0, u_{\max}],$$

reflecting limited treatment capacity, vaccine supply, or logistical constraints.

In addition, admissible controls are assumed to be *causal* (or nonanticipative): the control action at time t may depend only on information available up to time t, such as current or past infection levels, but not on future states of the system.

Objective functional. The planner seeks to minimize a cumulative cost functional of the form

$$J(u) = \int_{t_0}^{t_1} f\big(I(t), u(t), t\big)\, dt, \tag{2.6}$$

where the running cost f encodes trade-offs between public health outcomes and intervention effort. Typical choices include

$$f(I, u, t) = c_I I(t) + c_u u(t)^2,$$

where the first term penalizes high infection levels and the second term penalizes aggressive or costly interventions. Time-dependent costs may also be used to reflect changing priorities or seasonal effects.

Optimization problem. The epidemic control problem can thus be written as

$$\min_{u(\cdot) \in \mathcal{U}} J(u) \quad \text{subject to the SIR dynamics and given initial conditions.}$$

This is an infinite-dimensional optimization problem, as the decision variable is a function of time rather than a finite-dimensional vector.

Necessary conditions for optimality are typically derived using Pontryagin's Maximum Principle, which introduces adjoint variables associated with the state dynamics and yields a system of coupled state–costate equations. The resulting optimal control balances the marginal benefit of reducing infections against the marginal cost of intervention.

This example illustrates several defining features of optimal control in socio-technical systems: the explicit coupling between dynamics and decision variables, the role of constraints and causality, and the use of integral cost functionals to encode long-term objectives. Epidemic models of this form have been widely used to inform public health policy, resource allocation, and real-time decision-making during outbreaks.

2.4 Optimal Stopping

We next consider a discrete-time *optimal stopping problem*, in which a decision-maker observes the evolution of a stochastic process and must decide *when to stop* it so as to optimize an expected performance criterion. This formulation isolates the temporal tradeoff between waiting for additional information and committing to a terminal action, and it admits a clean dynamic programming characterization.

Let $\{X_k\}_{k\in\mathbb{Z}_{\geq 0}}$ be a discrete-time stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and adapted to the filtration $\{\mathcal{F}_k\}_{k\geq 0}$, where $\mathcal{F}_k$ represents the information available to the decision-maker up to time k. The process is assumed to be Markovian, with transition kernel

$$\mathbb{P}(X_{k+1} \in A \mid X_k = x), \qquad A \in \mathcal{B}(\mathcal{X}),$$

on a state space $\mathcal{X}$ (typically $\mathbb{R}^n$ or a finite set).

At each time k, after observing X_k, the decision-maker chooses one of two actions:

- *continue*, in which case the process evolves to X_{k+1} according to the transition kernel; or
- *stop*, in which case the process is terminated permanently and a terminal reward is realized.

A stopping rule is represented by a stopping time

$$\tau : \Omega \to \mathbb{Z}_{\geq 0} \cup \{\infty\},$$

such that $\{\tau = k\} \in \mathcal{F}_k$ for all k. Let $\mathcal{T}$ denote the set of all admissible stopping times with respect to $\{\mathcal{F}_k\}$.

Objective function. The objective is to maximize the expected reward accrued at the stopping time:

$$V(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x[\, G(X_\tau, \tau)\,], \tag{2.7}$$

where $G : \mathcal{X} \times \mathbb{Z}_{\geq 0} \to \mathbb{R}$ is a terminal reward function and $\mathbb{E}_x$ denotes expectation conditioned on the initial state $X_0 = x$.

A common and practically relevant specialization decomposes the reward into a terminal payoff and accumulated running costs:

Terminal reward

Continuation cost ∴ summed

$$G(X_\tau, \tau) = R(X_\tau) - \sum_{k=0}^{\tau-1} c(X_k), \tag{2.8}$$

where $c : \mathcal{X} \to \mathbb{R}_{\geq 0}$ is a per-stage continuation cost and $R : \mathcal{X} \to \mathbb{R}$ is the reward obtained upon stopping. This structure captures the tradeoff between waiting longer (incurring additional cost but potentially reaching a more favorable state) and stopping earlier.

The value function satisfies the Bellman optimality equation

$$V(x) = \max\{R(x),\ \mathbb{E}[V(X_{k+1}) \mid X_k = x]\}\,. \tag{2.9}$$

This recursion reflects the fundamental binary choice at each time step: stop immediately and receive reward $R(x)$, or continue and defer the stopping decision. Equation (2.9) induces a partition of the state space into:

- a *stopping region*, where $R(x) \geq \mathbb{E}[V(X_{k+1}) \mid X_k = x]$;
- a *continuation region*, where continuing yields higher expected value.

The optimal stopping time is the first time the process enters the stopping region.

Canonical example: timing of medical intervention. Let X_k represent a patient's health state evolving stochastically over discrete clinical visits $k = 0, 1, 2, \ldots$. Continuing observation provides additional diagnostic information but carries the risk of further health deterioration, while stopping corresponds to initiating an irreversible intervention such as surgery or aggressive treatment. The terminal reward $R(x)$ captures post-intervention health outcomes and quality of life, whereas the running cost $c(x)$ captures the physiological risk, discomfort, or mortality hazard associated with delaying action. The optimal stopping policy balances the value of information against the cost of waiting and typically induces a threshold structure: intervention is optimal once the health state falls below a critical level.

Canonical example: when to quit a project. Consider a project whose latent quality or viability evolves stochastically over discrete decision epochs $k = 0, 1, 2, \ldots$, represented by a state process X_k. At each stage, the decision-maker may either continue the project, incurring an operating cost $c(X_k)$ while observing new performance signals, or stop the project permanently and receive a terminal payoff $R(X_k)$, which may represent salvage value, opportunity cost recovery, or reallocation of resources. Continuing allows the agent to learn about the project's future potential, but also exposes the organization to further losses if the project is fundamentally unviable. The optimal policy is characterized by a stopping region in the state space: the project should be terminated once expected future returns fall below the continuation value. In many models, this policy takes the form of a threshold rule, whereby projects are abandoned when their estimated quality or belief state drops below a critical level.

In both cases, optimal stopping theory formalizes the trade-off between *information acquisition* and *irreversibility*, providing a normative benchmark for deciding when waiting is beneficial and when decisive action is optimal.

Behavioral Consideration Optimal stopping decisions are often made by humans and organizations, often without explicit modeling and analysis. Examples include deciding when to sell an asset, abandon a project, intervene medically, or terminate a failing strategy. Unlike the idealized decision-maker assumed in (2.9), human agents are subject to systematic behavioral biases, most notably the *sunk-cost fallacy*, loss aversion, and present bias. These effects lead decision-makers to continue suboptimal processes because of past investments rather than future expected value. From this perspective, optimal stopping theory provides not only a normative benchmark for rational decision-making, but also a diagnostic tool for identifying and quantifying departures from rational behavior in real-world systems.

3 Existence of Finite-Dimensional Optimization Problems

A central question in optimization theory is whether an optimization problem admits a solution. In general, the existence of a minimizer or maximizer cannot be taken for granted. Even in simple finite-dimensional settings, pathological behavior may arise if suitable regularity or compactness conditions are not imposed.

To illustrate the potential failure of existence, we begin with two elementary examples.

Example 1 (Unbounded Objective). Consider the optimization problem

$$\min_{x \in \mathbb{R}} x. \tag{3.1}$$

The objective function is linear and continuous, but the feasible set $\mathbb{R}$ is unbounded below. For any candidate solution x, one can always find a smaller value $x' < x$ such that $f(x') < f(x)$. Hence the objective decreases without bound, and no minimizer exists. Formally,

$$\inf_{x \in \mathbb{R}} x = -\infty,$$

and the problem has no solution. This example demonstrates that boundedness of the feasible set is essential for existence of minimizers.

Example 2 (Infimum Not Attained). Consider instead the problem

$$\min_{x \in \mathbb{R}} e^{-x}. \tag{3.2}$$

The objective function is continuous and bounded below, since $e^{-x} > 0$ for all $x \in \mathbb{R}$ and

$$\lim_{x \to +\infty} e^{-x} = 0.$$

Thus,

$$\inf_{x \in \mathbb{R}} e^{-x} = 0.$$

However, there exists no $x \in \mathbb{R}$ such that $e^{-x} = 0$. Consequently, although the objective admits a finite lower bound, the infimum is not attained, and the optimization problem has no minimizer. This example shows that boundedness of the objective alone is insufficient for existence.

Remark (Minimum versus Infimum). These examples highlight the crucial distinction between the *minimum* and the *infimum* of a function. The minimum of a function f over a set C is a value $f(x^*)$ achieved at some point $x^* \in C$, whereas the infimum is merely the greatest lower bound of the set $\{f(x) : x \in C\}$ and need not be attained.

As a consequence, expressions such as

$$\min_{x \in \mathbb{R}} e^{-x} = 0 \tag{3.3}$$

are mathematically incorrect, since no minimizer exists. The correct statement is

$$\inf_{x \in \mathbb{R}} e^{-x} = 0.$$

Failure to distinguish between these notions can lead to conceptual and technical errors in optimization analysis.

3.1 Weierstrass Extreme Value Theorem

The preceding examples motivate the need for conditions that guarantee existence of solutions. In finite-dimensional spaces, such guarantees are provided by compactness and continuity.

Theorem 3.1 (Weierstrass Extreme Value Theorem)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous function, and let $C \subseteq \mathbb{R}^n$ be compact, i.e., closed and bounded. Then there exist points $x^*, x^\dagger \in C$ such that*

$$f(x^*) = \min_{x \in C} f(x), \qquad f(x^\dagger) = \max_{x \in C} f(x). \tag{3.4}$$

Proof. We prove existence of a minimizer; the maximizer then follows by applying the same argument to $-f$.

Let

$$m := \inf_{x \in C} f(x).$$

By the definition of infimum, for each $k \in \mathbb{N}$ there exists $x_k \in C$ such that

$$f(x_k) \leq m + \frac{1}{k}.$$

Thus $\{x_k\}$ is a minimizing sequence in C.

Since C is compact, every sequence in C has a convergent subsequence with limit in C. Hence there exist indices $k_j \to \infty$ and a point $x^* \in C$ such that

$$x_{k_j} \to x^* \quad \text{as } j \to \infty.$$

Because f is continuous, we have

$$f(x_{k_j}) \to f(x^*) \quad \text{as } j \to \infty.$$

On the other hand, for all j,

$$m \leq f(x_{k_j}) \leq m + \frac{1}{k_j}.$$

Letting $j \to \infty$ and using $k_j \to \infty$ yields

$$\lim_{j\to\infty} f(x_{k_j}) = m.$$

Therefore $f(x^*) = m$, and since $x^* \in C$, it follows that

$$f(x^*) = \min_{x\in C} f(x).$$

To obtain a maximizer, apply the previous argument to the continuous function $-f$ on the compact set C. Then there exists $x^\dagger \in C$ such that

$$-f(x^\dagger) = \min_{x\in C}(-f(x)),$$

which is equivalent to $f(x^\dagger) = \max_{x\in C} f(x)$. □

The Weierstrass theorem establishes that, in finite-dimensional spaces, continuity of the objective function and compactness of the feasible set are sufficient to ensure existence of both minimizers and maximizers. The boundedness of C prevents escape to infinity, while closedness ensures that limiting points of minimizing sequences remain feasible.

This result underpins much of classical optimization theory. Many existence results for constrained optimization problems are obtained by reformulating the feasible set to be compact or by proving that minimizing sequences are precompact. When compactness fails, existence may still hold, but additional arguments are required. In infinite-dimensional spaces, closed and bounded sets need not be compact, and existence of minimizers becomes a substantially more delicate issue.

For minimization problems, continuity of the objective function can be relaxed. In particular, it is sufficient to require a one-sided regularity condition that prevents the objective from "jumping downward" at limit points.

Definition 3.2 (Lower semicontinuity)**.** A function $f : C \to \mathbb{R} \cup \{+\infty\}$ is said to be *lower semicontinuous* if for every sequence $\{x_n\} \subset C$ such that $x_n \to x \in C$, we have

$$f(x) \leq \liminf_{n\to\infty} f(x_n). \tag{3.5}$$

Equivalently, f is lower semicontinuous if and only if all its sublevel sets

$$\{x \in C : f(x) \leq \alpha\}, \qquad \alpha \in \mathbb{R},$$

are closed subsets of C.

Theorem 3.3. *If C is compact and $f : C \to \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous, then f attains its minimum on C, i.e., there exists $x^* \in C$ such that*

$$f(x^*) = \min_{x \in C} f(x).$$

Proof. Define $m := \inf_{x \in C} f(x)$. If $m = +\infty$, then $f(x) = +\infty$ for all $x \in C$, and any $x^* \in C$ trivially attains the minimum. We therefore assume $m < +\infty$.

By the definition of infimum, for each $k \in \mathbb{N}$ there exists $x_k \in C$ such that $f(x_k) \leq m + \frac{1}{k}$. Thus $\{x_k\}$ is a minimizing sequence in C.

Since C is compact, there exists a subsequence $\{x_{k_j}\}$ and a point $x^* \in C$ such that $x_{k_j} \to x^*$. By lower semicontinuity of f, we have $f(x^*) \leq \liminf_{j \to \infty} f(x_{k_j})$. On the other hand, from the construction of the minimizing sequence, $m \leq f(x_{k_j}) \leq m + \frac{1}{k_j}$. Taking the limit inferior as $j \to \infty$ yields $\liminf_{j \to \infty} f(x_{k_j}) = m$. Combining the inequalities gives $f(x^*) \leq m$. Since m is the infimum of f over C, we also have $f(x^*) \geq m$, and therefore $f(x^*) = m$. Hence x^* attains the minimum of f on C. $\square$

Lower semicontinuity is a weaker condition than continuity. A function may fail to be continuous, or even be discontinuous on a dense set, and still be lower semicontinuous. Such functions may not admit a maximum, but can nevertheless admit a minimum.

This asymmetry reflects the directional nature of optimization. For minimization problems, lower semicontinuity prevents the objective from exhibiting downward jumps that would destroy attainment of the infimum. By contrast, existence of maximizers requires upper semicontinuity. In particular, a function that is lower semicontinuous need not attain a maximum on a compact set.

As a result, lower semicontinuity is the essential regularity condition for existence of solutions in minimization problems, especially in variational analysis, convex optimization, and infinite-dimensional optimization.

3.2 Coercive Functions

The Weierstrass extreme value theorem does not always apply. In particular, if the feasible set is not compact or if the objective function fails to grow sufficiently at infinity, the infimum of an optimization problem may not be attained. A classical and widely used sufficient condition ensuring existence of solutions on unbounded domains is *coercivity*. Throughout, unless otherwise stated, $|\cdot|$ denotes the Euclidean norm on $\mathbb{R}^n$.

Definition 3.4 (Coercivity)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *coercive* if

$$\lim_{|x|\to\infty} f(x) = +\infty. \tag{3.6}$$

Equivalently, for every $c \in \mathbb{R}$ there exists $R > 0$ such that $f(x) > c$ whenever $|x| > R$.

Coercivity ensures that the objective function grows without bound as the decision variable escapes to infinity, thereby preventing minimizing sequences from diverging.

Lemma 3.5 (Compactness of sublevel sets)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and coercive. Then for every $c \in \mathbb{R}$, the sublevel set*

$$L_c := \{x \in \mathbb{R}^n : f(x) \leq c\} \tag{3.7}$$

is compact.

Proof. By coercivity, there exists $R > 0$ such that $f(x) > c$ for all $|x| > R$. Hence $L_c \subseteq \overline{B}_R(0)$ and is bounded. Continuity of f implies that L_c is closed as the preimage of $(-\infty, c]$ under a continuous mapping. Since closed and bounded sets are compact in $\mathbb{R}^n$, the result follows. □

Theorem 3.6 (Existence of a minimizer for coercive functions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and coercive. Then the unconstrained optimization problem*

$$\min_{x\in\mathbb{R}^n} f(x) \tag{3.8}$$

admits at least one minimizer.

Proof. Let $m := \inf_{x\in\mathbb{R}^n} f(x)$ and fix any $c > m$. The sublevel set L_c is nonempty and compact by Lemma 3.5. By the Weierstrass extreme value theorem, there exists $x^* \in L_c$ such that

$$f(x^*) = \min_{x\in L_c} f(x).$$

Since any $x \notin L_c$ satisfies $f(x) > c > m$, such points cannot minimize f over $\mathbb{R}^n$. Therefore, x^* is a global minimizer of f. □

The coercivity argument extends naturally to constrained optimization when compactness of a suitable level set can be established.

Theorem 3.7 (Existence via compact level sets)**.** *Let $D \subseteq \mathbb{R}^n$ and let $f : D \to \mathbb{R}$ be continuous. Suppose there exists $c \in \mathbb{R}$ such that the sublevel set*

$$L_c := \{x \in D : f(x) \leq c\} \tag{3.9}$$

is nonempty and compact. Then the optimization problem

$$\min_{x \in D} f(x) \tag{3.10}$$

admits at least one minimizer.

Proof. Since L_c is compact and f is continuous, the Weierstrass extreme value theorem implies that f attains its minimum on L_c. Any minimizer of f over D must lie in L_c, and hence the minimum is attained on D. □

4 Generalized Weierstrass Theorem for Optimization in Normed Spaces

In finite-dimensional spaces such as $\mathbb{R}^n$, existence of minimizers and maximizers is guaranteed under mild assumptions, most notably continuity of the objective function and compactness of the feasible set. This is the content of the classical Weierstrass extreme value theorem.

When optimization problems are posed in more general vector spaces—such as function spaces arising in control, signal processing, or variational problems—the same ideas continue to apply, but the underlying assumptions must be stated more carefully. In particular, continuity can be weakened to semicontinuity, and compactness must be understood in a topological sense rather than as closedness and boundedness.

4.1 Background Concepts

We begin by recalling the key notions needed to formulate a generalized existence theorem.

Normed linear spaces. A *normed linear space* $(X, \|\cdot\|)$ is a vector space equipped with a norm that induces a metric and hence a notion of convergence. A sequence $\{x_n\} \subset X$ is said to converge to $x \in X$ if

$$\|x_n - x\| \to 0 \quad \text{as } n \to \infty.$$

Compactness. A set $K \subset X$ is said to be *compact* if every sequence $\{x_n\} \subset K$ admits a subsequence that converges to an element of K. This property ensures that minimizing or maximizing sequences do not "escape" the feasible set.

In $\mathbb{R}^n$, compactness is equivalent to being closed and bounded (Heine–Borel theorem). In infinite-dimensional normed spaces, however, closed and bounded sets need not be compact, making existence results more subtle.

Upper semicontinuity. A function $f : X \to \mathbb{R}$ is said to be *upper semicontinuous* at a point $x_0 \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f(x) \leq f(x_0) + \varepsilon \quad \text{for all } x \in B_\delta(x_0), \tag{4.1}$$

where $B_\delta(x_0)$ denotes the open ball of radius δ centered at x_0.

Intuitively, upper semicontinuity prevents the function from exhibiting upward jumps at limit points.

Lower semicontinuity. Similarly, a function $f : X \to \mathbb{R}$ is *lower semicontinuous* at x_0 if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f(x) \geq f(x_0) - \varepsilon \quad \text{for all } x \in B_\delta(x_0).$$

Lower semicontinuity prevents downward jumps and is the natural regularity condition for minimization problems.

Equivalent sequential characterizations are often useful: f is lower semicontinuous if and only if

$$f(x) \leq \liminf_{n\to\infty} f(x_n) \quad \text{whenever } x_n \to x,$$

and upper semicontinuous if and only if

$$f(x) \geq \limsup_{n\to\infty} f(x_n).$$

4.2 Generalized Weierstrass Theorem

We are now ready to state the generalized version of the Weierstrass theorem.

Theorem 4.1 (Generalized Weierstrass Extreme Value Theorem)**.** *Let X be a normed linear space and let $K \subset X$ be compact.*

- *If $f : K \to \mathbb{R}$ is upper semicontinuous, then f attains its maximum on K, i.e., there exists $x^\dagger \in K$ such that*

$$f(x^\dagger) = \max_{x \in K} f(x).$$

- *If $f : K \to \mathbb{R}$ is lower semicontinuous, then f attains its minimum on K, i.e., there exists $x^* \in K$ such that*

$$f(x^*) = \min_{x \in K} f(x).$$

This theorem generalizes the classical finite-dimensional result in two important ways. First, it applies to arbitrary normed spaces rather than only to $\mathbb{R}^n$. Second, it shows that full continuity of the objective function is not required: one-sided semicontinuity suffices, depending on whether a maximization or minimization problem is considered.

The asymmetry between maxima and minima reflects the directional nature of optimization. Upper semicontinuity is essential for guaranteeing existence of maximizers, while lower semicontinuity is essential for minimizers.

In infinite-dimensional spaces, compactness is a strong assumption and often fails for natural feasible sets. Much of modern optimization theory focuses on recovering existence results by replacing compactness with weaker conditions, such as coercivity, reflexivity, or weak compactness, together with appropriate semicontinuity assumptions.

5 Local and Global Optimality

We consider the optimization problem

$$\min_{x \in D} f(x), \tag{5.1}$$

where $D \subseteq \mathbb{R}^n$ is the feasible set and $f : D \to \mathbb{R}$ is the objective function. The goal is to find points in D at which f takes its smallest possible value.

5.1 Global Optimality

We begin with the notion of optimality over the entire feasible set.

Definition 5.1 (Optimal value)**.** The scalar

$$f^* := \inf_{x \in D} f(x) \tag{5.2}$$

is called the *optimal value* of the problem. It represents the smallest value that the objective function can achieve over D, whether or not this value is actually attained.

Definition 5.2 (Global minimizer (optimal point))**.** A point $x^* \in D$ is called a *global minimizer* (or simply an *optimal point*) if

$$f(x^*) = f^*. \tag{5.3}$$

In this case, the optimal value is said to be *attained*.

Definition 5.3 (Set of global minimizers)**.** The set of all global minimizers is denoted by

$$X_{\text{opt}} := \{x \in D : f(x) = f^*\}. \tag{5.4}$$

This set may contain a single point, multiple points, or be empty if the infimum is not attained.

5.2 Local Optimality

In many problems, especially nonconvex ones, it is useful to distinguish between points that are optimal globally and those that are optimal only in a neighborhood.

Definition 5.4 (Local minimizer)**.** A point $x^* \in D$ is called a *local minimizer* of f if there exists $\delta > 0$ such that

$$f(x^*) \leq f(x), \quad \forall x \in B_\delta(x^*) \cap D, \tag{5.5}$$

where $B_\delta(x^*)$ denotes the open ball of radius δ centered at x^*.

Intuitively, a local minimizer is a point at which the objective function cannot be decreased by making small feasible perturbations, even though smaller values of f may exist elsewhere in D.

Every global minimizer is necessarily a local minimizer, since a point that is optimal over the entire feasible set is also optimal in any neighborhood.

However, the converse is not true in general: a local minimizer need not be a global minimizer. Local minima are common in nonconvex optimization problems, where the objective function may have many valleys of varying depth.

If both the feasible set D and the objective function f are convex, then every local minimizer is also a global minimizer. This property plays a central role in convex optimization and explains why convex problems are significantly easier to analyze and solve than general nonconvex problems.

References